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Vero: An Open RL Recipe for General Visual Reasoning

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Models Data Code Project Page

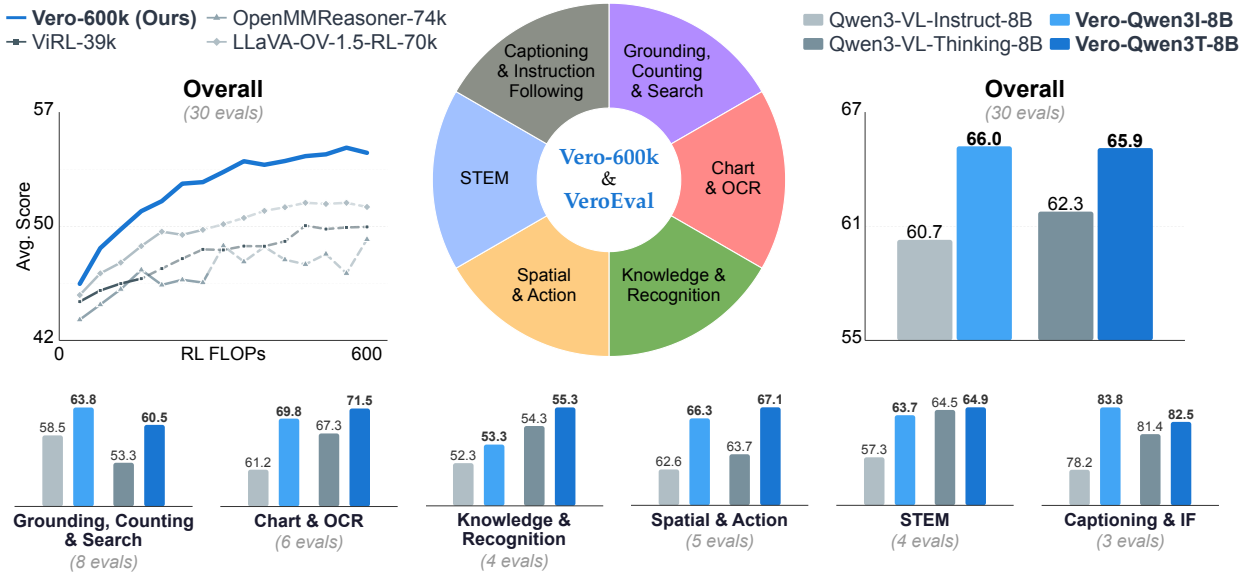


Figure 1. Vero achieves state-of-the-art performance across six task categories using a fully open RL recipe. Top left: Training curves versus RL FLOPs for Vero-600K compared with existing open RL datasets, all finetuned from Qwen2.5-VL-7B-Instruct; dashed lines indicate training beyond one epoch. **Top center:** Summary of the broad task categories targeted in Vero-600K and VeroEval. **Top right:** Vero compared to Qwen3-VL-Instruct and Qwen3-VL-Thinking overall on the 30 benchmarks in VeroEval. **Bottom row:** Same as top right, but displaying per-category scores.

Abstract

What does it take to build a visual reasoner that works across charts, science, spatial understanding, and open-ended tasks? The strongest vision-language models (VLMs) show such broad visual reasoning is within reach, but the recipe behind them remains unclear, locked behind proprietary reinforcement learning (RL) pipelines with non-public data. We introduce Vero, a family of *fully open* VLMs that matches or exceeds existing open-weight models across diverse visual reasoning tasks. We scale RL data and rewards across six broad task categories, constructing Vero-600K, a 600K-sample dataset from 59 datasets, and designing task-routed rewards that handle heterogeneous answer formats. Vero achieves state-of-the-art performance, improving over four base models by 3.6–5.3 points on average across VeroEval, our suite of 30 challenging benchmarks. Starting from Qwen3-VL-8B-Instruct, Vero outperforms Qwen3-VL-8B-Thinking on 23 of 30 benchmarks without additional proprietary thinking data. When trained from the same base model, Vero-600K exceeds existing RL datasets across task categories. Systematic ablations reveal that different task categories elicit qualitatively distinct reasoning patterns that transfer poorly in isolation, suggesting that broad data coverage is the primary driver of strong RL scaling. All data, code, and models are released.

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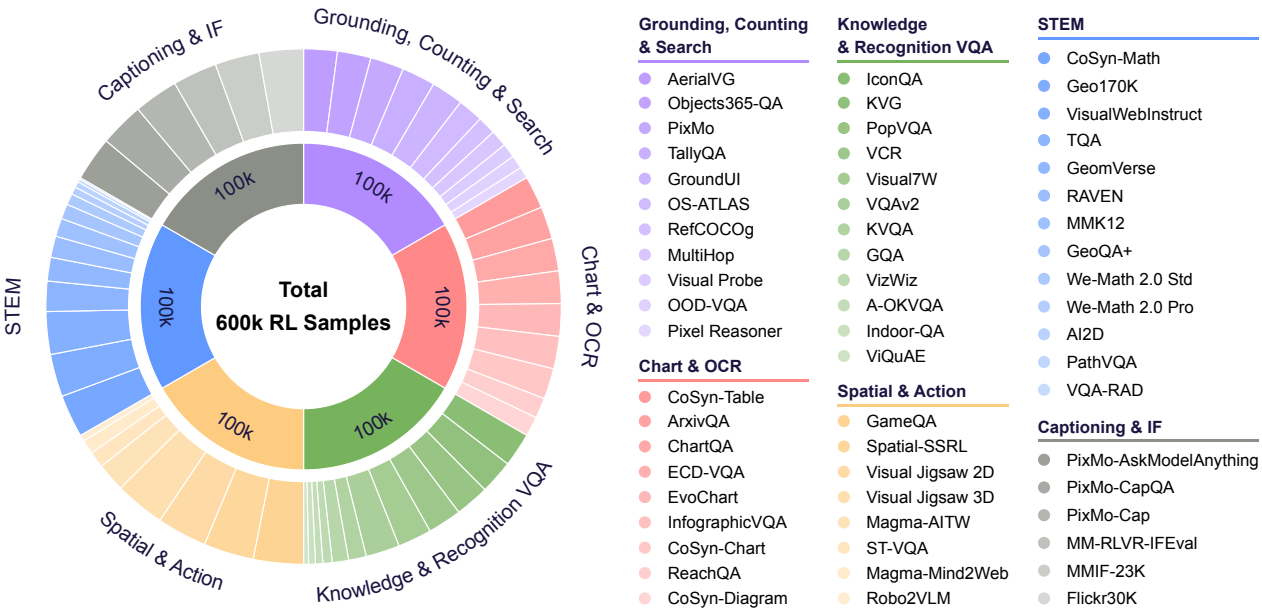


Figure 2. Composition of Vero-600K. The inner ring shows six task categories, each allocated 100K samples (600K total), and the outer ring shows their 59 constituent datasets. The categories represent real-world use cases and cover distinct visual reasoning capabilities (Sections 5 and 7). Categories are uniformly sampled to balance learning across tasks.

1 Introduction

Vision-language models (VLMs) are increasingly expected to reason across a wide range of visual tasks, from chart and scientific interpretation to spatial understanding and open-ended questions. Reinforcement learning (RL) has emerged as a key driver of this progress, with methods such as PPO (Schulman et al., 2017) and GRPO (Shao et al., 2024) enabling models to learn from their own generations through reward signals. Recent multimodal models such as GPT-5 (Singh et al., 2025), Qwen3-VL (Bai et al., 2025a), and Kimi K2.5 (Kimi Team et al., 2026) demonstrate that RL drives substantial improvement in multimodal reasoning.

Yet the strongest existing visual reasoning models are products of proprietary RL pipelines with non-public data and undisclosed reward designs. Models such as Qwen3-VL (Bai et al., 2025a) release weights and are widely adopted, but do not release RL training code or datasets. Accompanying technical reports often omit detailed ablations of design choices, making it difficult to systematically study what drives performance. Meanwhile, fully open efforts such as OpenMMReasoner (Zhang et al., 2026) and VL-Rethinker (Wang et al., 2025) focus primarily on visual math, covering only a narrow subset of visual tasks. However, as we show in Sections 5 and 7, training on a single task category does not generalize to other visual capabilities, in both task performance and chain-of-thought behavior. More broadly, applying RL across heterogeneous visual reasoning tasks is challenging, as diverse task mixtures induce interference, weak transfer, and optimization imbalance unless the training distribution and rewards are carefully designed (Teh et al., 2017; Schaul et al., 2019; Hessel et al., 2019). This leaves a central question: *what does it take to train a broadly capable visual reasoner?*

We show that a single-stage RL recipe with diverse and high quality data suffices. We introduce **Vero**, a family of fully open VLMs trained with RL on top of existing models to perform strongly across diverse visual tasks. Our recipe centers on careful dataset selection, sample filtering, and task-balanced mixing: we build **Vero-600K**, a 600K-sample training set from 59 datasets spanning six core task categories (Chart & OCR; STEM; Spatial & Action; Knowledge & Recognition; Grounding, Counting & Search; and Captioning & Instruction Following), and pair it with task-routed reward functions. No additional warm start, no staged RL, and no proprietary data. Alongside the training data, we assemble **VeroEval**, a comprehensive evaluation suite of 30 benchmarks spanning all six categories. Figure 2 summarizes the composition of **Vero-600K**.

Through systematic ablations of dataset selection, sample filtering, mixture strategies, and reward design, we find that data diversity is the critical ingredient. Different task categories elicit qualitatively distinct reasoning

patterns that transfer poorly in isolation: for example, STEM tasks trigger elevated backtracking while grounding tasks suppress introspective behaviors in favor of directed visual search. Broad task coverage is therefore essential for producing a generally capable model. We additionally find that (1) uniform mixture weighting across task categories outperforms schemes based on accuracy, reasoning length, or dataset size; (2) multi-task training necessitates an expressive, task-routed reward design; and (3) including open-ended tasks is necessary to preserve visual chat ability during RL.

Vero achieves state-of-the-art overall performance among 8B VLMs (Figure 1). Training on four different base models yields consistent improvements of +3.6 to +5.3 points averaged over 30 benchmarks. **Vero-Qwen3T-8B**, trained from Qwen3-VL-8B-Thinking, outperforms Qwen3-VL-8B-Thinking on 24 of 30 benchmarks, with gains of +6.9 on CharXiv$_{Reason}$, and +5.8 on CountQA. **Vero-Qwen3I-8B**, trained from Qwen3-VL-8B-Instruct, outperforms Qwen3-VL-8B-Thinking on 23 of 30 benchmarks. **Vero-MiMo-7B**, trained on MiMo-VL-7B-SFT (Yue et al., 2025b) with our open recipe, surpasses MiMo-VL-7B-RL, which starts from the same initial model but uses a proprietary RL recipe with non-public data. We release all data, code, and models to facilitate future research.

2 Related Work

Vision-language models. Vision-language models excel on multimodal tasks, including proprietary systems such as GPT-5 (Singh et al., 2025) and Gemini (Team et al., 2023, 2024; Comanici et al., 2025), open-weight families such as Qwen (Bai et al., 2025b,a), GLM (Hong et al., 2025), and Kimi (Kimi Team, 2025), and fully open releases of data, code, and weights such as Molmo (Deitke et al., 2025; Clark et al., 2026) and LLaVA (Liu et al., 2023; An et al., 2025). These models are expected to handle a wide range of tasks. While little is publicly known about proprietary model post-training, recent open-weight models have explored techniques such as RL with curriculum sampling (Hong et al., 2025) and mixed on-policy RL (Yue et al., 2025b), yet the factors that drive their performance across diverse tasks remain unclear. Our work targets this gap by providing a fully open multi-domain RL recipe for general visual understanding.

Reasoning and thinking for VLMs. Chain-of-thought reasoning enables models to use additional test-time compute through step-by-step problem decomposition (Wei et al., 2022; Zhang et al., 2024). The two dominant approaches for training reasoning models are distillation, where a strong teacher generates reasoning traces for supervised fine-tuning (Xu et al., 2025; Yao et al., 2025; Sarch et al., 2025), and reinforcement learning, which optimizes against outcome-based rewards without requiring a fixed teacher (DeepSeek-AI et al., 2025). Recent works apply RL to visual reasoning (Yu et al., 2025a; Wang et al., 2025; Zhang et al., 2026), but primarily in narrow domains, leaving the effect of RL-trained reasoning on broad visual understanding underexplored. We show that RL with careful reward and data design consistently outperforms narrowly trained baselines across diverse visual task categories.

RL recipes and training data design for VLMs. Several works provide recipes for RL-based visual reasoning training. OpenMMReasoner (Zhang et al., 2026) combines teacher distillation and GSPO (Zheng et al., 2025a) over multimodal reasoning benchmarks, VL-Rethinker (Wang et al., 2025) addresses training instability via selective sample replay and forced rethinking, and Perception-R1 (Yu et al., 2025a) designs discriminative rewards for perceptual tasks such as grounding and counting. These efforts primarily target visual math or narrow perceptual domains and provide only limited ablations of dataset selection, sample filtering, and reward design. Our recipe centers on **Vero-600K**, which spans six task categories with 600K datapoints from 59 datasets, includes a routed reward system, and provides systematic ablations of design choices, all released publicly to support open VLM research.

3 Vero

3.1 Task Categories

We consider the problem of training a Vision-Language Model (VLM) π_θ via reinforcement learning to maximize expected reward across a diverse set of visual reasoning tasks. Given a visual input v (an image or

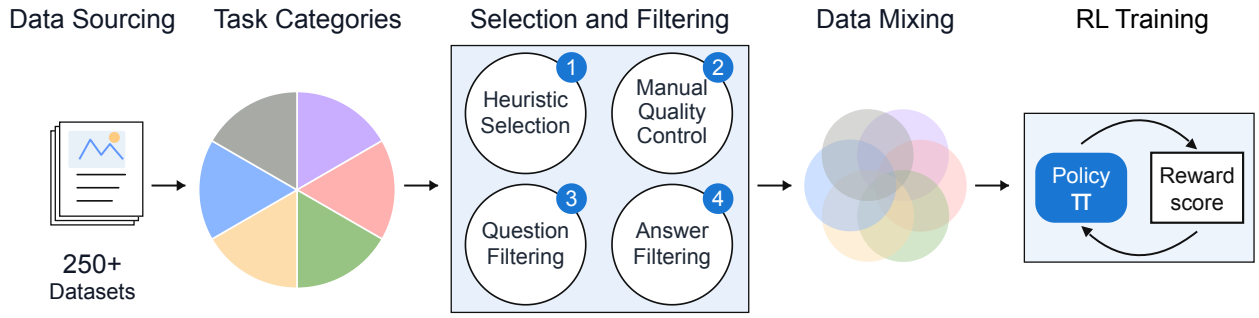


Figure 3. Vero-600K data curation pipeline. Starting from over 250 candidate datasets, we assign each to one of six task categories and apply multi-stage selection and filtering: heuristic screening (size, resolution, answer format), manual quality control, LLM-based question filtering for ambiguity and verifiability, and answer filtering for stable reward computation. The retained data are combined into a uniformly weighted mixture across task categories and used for on-policy RL training with task-routed rewards.

set of images) and a text query q, the model generates a structured response $y = (z, a) \sim \pi_\theta(\cdot \mid v, q)$, where z denotes the reasoning or thinking content and a denotes the final answer. We verify the final answer a against the ground-truth answer y^*. The RL training objective is:

$$\max_\theta \ \mathbb{E}_{(v,q,y^*)\sim\mathcal{D}} \ \mathbb{E}_{(z,a)\sim\pi_\theta(\cdot|v,q)} \left[R(a, y^*)\right], \tag{1}$$

where $\mathcal{D}$ is the training data distribution. A central challenge is constructing $\mathcal{D}$ to span a broad range of visual reasoning capabilities, so that the resulting policy generalizes across diverse tasks.

Figure 2 provides an overview of our training data composition. We organize our training data into six task categories, each targeting a distinct visual reasoning capability. This taxonomy is motivated by two observations. First, we find empirically (Section 5 and Section 7) that training on any single category fails to transfer reliably to others and elicits distinct chain-of-thought behaviors, suggesting that these categories exercise different reasoning strategies and skills. Second, while existing VLM evaluation frameworks organize benchmarks along similar axes, e.g., Qwen2.5-VL (Bai et al., 2025b) separates document understanding, mathematical reasoning, and grounding, and Kimi K2.5 (Kimi Team et al., 2026) distinguishes reasoning from perception, these categorizations are typically adopted by convention rather than validated empirically, and they are designed for evaluation rather than training. Our taxonomy refines and extends these axes to cover a broader set of visual reasoning tasks, and we validate its effectiveness for multi-task RL (Section 5).

Concretely, we define six categories: **STEM** (13 datasets) covers mathematical diagram reasoning, scientific figure interpretation, and medical image understanding, with answers that are typically numeric or symbolic. **Spatial & Action** (8 datasets) targets embodied reasoning, UI navigation, and 3D spatial understanding, requiring reasoning about spatial transformations and action sequences. **Knowledge & Recognition** (12 datasets) spans visual question answering that combines object, scene, and entity recognition with external or commonsense knowledge. **Chart & OCR** (9 datasets) focuses on extracting and reasoning over structured information in documents, charts, tables, and infographics. **Grounding, Counting & Search** (11 datasets) requires spatially localizing objects via bounding boxes, counting entity instances, and searching among visual distractors. **Captioning & Instruction Following** (6 datasets) encompasses open-ended image description and following prompt instructions.

3.2 Vero-600K: Sourcing, Filtering, and Mixtures of Broad Tasks

We construct **Vero-600K**, a multi-task RL training set of 600K samples from 59 datasets spanning six task categories (Section 3.1). Figure 3 summarizes the **Vero-600K** data curation pipeline, and Figure 4 shows representative examples from all six task categories.

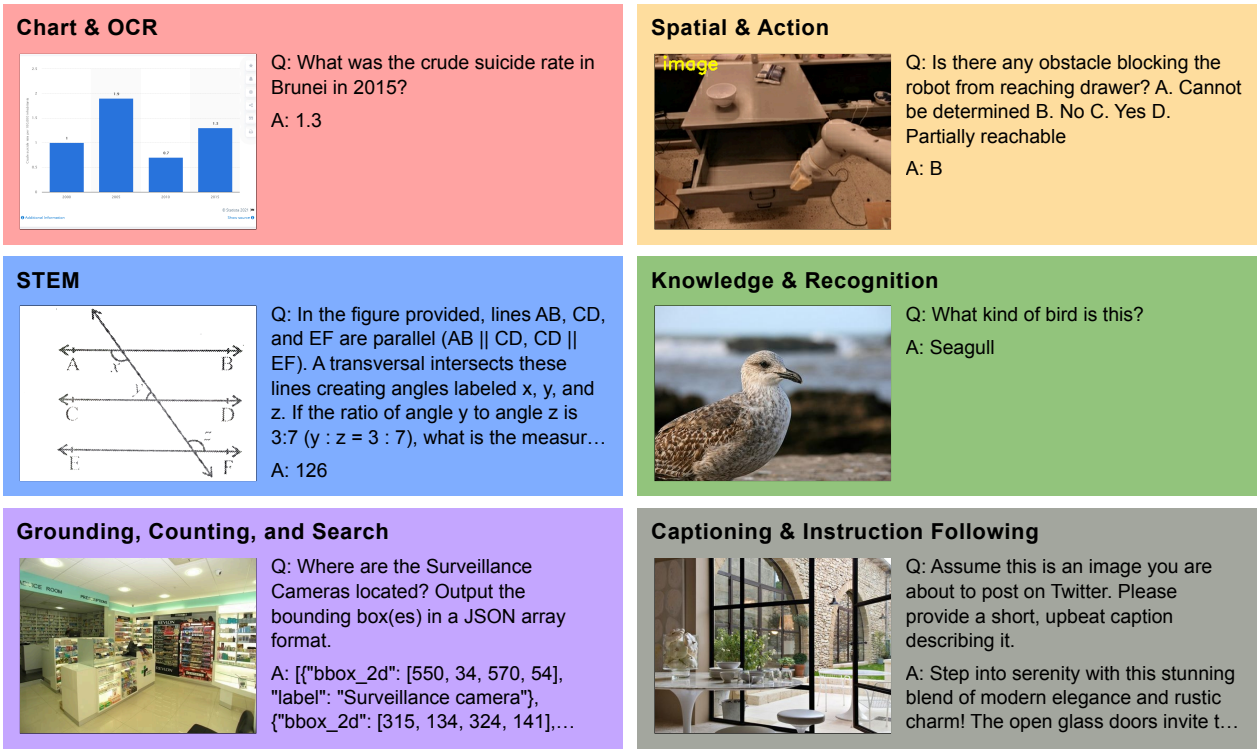


Figure 4. Examples from each task category illustrate the breadth of Vero-600K. We show representative samples of our training data from the six categories, highlighting the diversity of visual inputs, question formats, and answer types covered by our training set.

Step 1. Dataset sourcing and selection. We start from over 250 candidate datasets drawn from instruction-tuning and RL collections (e.g., FineVision (Wiedmann et al., 2025)) and recently released task-specific sources (e.g., Visual Jigsaw (Wu et al., 2025a)), then apply dataset-level filtering. Each dataset is assigned to the task category that best reflects its primary skill, based on manual inspection and its utility in prior work.

Heuristic selection. We discard datasets with fewer than 1K examples, average image resolution below 200K pixels (retaining five low-resolution datasets for question quality), or binary questions to mitigate guessing.

Manual selection. For each candidate, we inspect ∼50 examples against three criteria: **correctness** (<5% annotation error rate in image–question–answer triples), **unambiguity** (each question admits a single verifiable answer), and **verifiability** (the answer format is compatible with our reward functions). Of ∼100 datasets passing heuristic screening, 59 were retained. For a small number of datasets, we additionally rewrite questions to fix prompt clarity (e.g., GameQA, Magma) or drop high-error subsets.

Figure D1 (Appendix) compares dataset selection strategies and shows our filtering has significant gains compared to taking a random subset from the candidate pool or sampling from the STEM or Chart subsets of the FineVision (Wiedmann et al., 2025) training set. On STEM, our selected mixture achieves an average benchmark gain of +4.6, compared to +2.9 for FineVision and −0.2 for random sampling from all candidates. On Chart & OCR, the gains are +3.4, +1.9, and +2.5, respectively.

Step 2. Data filtering. After dataset-level filtering, many individual examples remain ambiguous, unanswerable, or incompatible with our rewards. We apply additional steps to filter individual prompts.

Question filtering. We use Qwen3-VL-235B-A22B-Instruct (Bai et al., 2025a) to remove ambiguous, image-irrelevant, or unverifiable questions. The model scores each datapoint on five criteria: (1) *relevance*, whether the image depicts what the question refers to; (2) *ambiguity*, whether the question is too vague or not a

	Chart & OCR	STEM	Spatial & Action	Knowl. & Recog.	Grnd., Cnt. & Search
Unfiltered	60.0	45.4	56.3	62.5	**54.7**
Q. Filtering	**60.1**	43.6	**58.2**	63.0	54.1
A. Canonic.	59.9	**45.5**	–	**64.6**	–

Table 1. Filtering generally helps remove ambiguous samples from noisy datasets. Effect of question filtering and answer filtering on Qwen2.5-VL-7B-Instruct. This table shows category average scores on **VeroEval**.

	Chart & OCR	STEM	Spatial & Action	Knowl. & Recog.	Grnd., Cnt. & Search	Bench. Avg.
equal ratios	**+8.6**	+6.2	**+5.6**	+1.8	+5.6	**+5.8**
ratio $\propto (1 - \text{acc.})^{\alpha}$	+6.8	**+6.5**	+4.3	**+2.4**	+5.2	+5.2
ratio $\propto \text{area}^{\alpha}$	+7.0	+5.3	+4.1	+1.4	**+6.2**	+5.2
ratio $\propto \text{length}^{\alpha}$	+7.5	+6.4	+4.5	+1.7	+3.8	+4.8
w/o Knowl. & Recog.	+6.4	**+6.5**	+4.8	+1.9	+4.7	+4.9

Table 2. Equal task ratios perform best overall. Weighting schemes: equal ratios (uniform), difficulty-weighted by inverse accuracy, image-area-weighted by mean input resolution, reasoning-length-weighted by mean chain-of-thought length, and ablation dropping Knowledge & Recognition. Values are absolute score changes (Δ) on **VeroEval** when using Qwen3-VL-8B-Instruct as the initial model for RL.

genuine question; (3) *language*, whether the question is in English; (4) *verifiability*, whether a single objectively correct answer can be derived from visible content; and (5) *numeric precision*, whether the required precision is visually unambiguous. Any triggered criterion removes the datapoint. We provide details in Appendix A.3.

Answer filtering. We normalize ground-truth answers using text-only Qwen3-235B-A22B-Instruct (Bai et al., 2025a) to ensure stable reward computation. Numeric answers are stripped of units and currency symbols, converted to decimal form, and evaluated as expressions. Samples with unsupported notation are filtered. Multiple-choice answers are normalized to a single canonical letter. Except for the captioning and instruction following task category, samples with multi-value answers, non-reducible symbolic expressions, or ambiguous descriptions requiring semantic matching are removed. A full list of answer filtering rules is in the Appendix A.4. We do not apply it to Spatial & Action or Grounding tasks, since answers are already standardized for all datasets in these categories.

Effects of filtering are mixed across categories (Table 1): question filtering yields a clear gain on Spatial & Action (+1.9 pts) but slightly hurts Grounding, Counting & Search (−0.6 pts), while answer filtering substantially improves Knowledge & Recognition (+2.1 pts) but is flat or marginally negative on other categories. Despite this variance, we apply both steps to all applicable task categories, as they generally help remove ambiguous samples from noisy datasets and the largest gains outweigh the small regressions.

Step 3. Data mixtures. In our multi-task RL setting, the task category sampling distribution governs how training signal is allocated across skills. We investigate four task category weighting schemes (uniform, difficulty-weighted, dataset-size-weighted, reasoning-length-weighted), where the amount of samples per batch is determined by a ratio proportional to the metric (e.g., difficulty). Uniform sampling achieves the highest benchmark average gain (+5.8 pts over the base model), outperforming alternative schemes. Alternatives yield gains on individual categories but at the cost of others (Table 2). We use uniform task category weighting, as it achieves the best overall performance.

VeroEval evaluation suite. We introduce **VeroEval**, a challenging evaluation suite for broad visual reasoning. We curate a suite of 30 benchmarks spanning the six visual reasoning categories defined in Section 3, with three to eight benchmarks per category. We select benchmarks according to three criteria: (i) *difficulty*: we favor benchmarks on which current frontier models have room for improvement, while retaining established benchmarks (e.g., ChartQA, ScreenSpot) for comparability with prior work; (ii) *annotation quality*: we include only benchmarks with well-defined evaluation protocols and reliable ground-truth labels; and (iii) *intra-category diversity*: within each category we include benchmarks that test complementary sub-skills (e.g., within Chart & OCR: chart reasoning, infographic understanding, and scientific figure interpretation). The full benchmark list appears in Appendix Table A2.

3.3 Training Vero with Reinforcement Learning

Algorithmic details. At its core, RL maximizes the expected reward of the model's response y given a visual input v and query q. Our RL algorithm builds on Group Relative Policy Optimization (GRPO) (Shao et al., 2024) and integrates algorithmic advances from GSPO (Zheng et al., 2025a), among others (Yu et al., 2025b).

GSPO (Zheng et al., 2025a) replaces the independent per-token importance ratios of GRPO with a *sequence-level* ratio. For each response y_i in a group of G rollouts, the sequence-average log-probability difference $\bar{\Delta}_i = \frac{1}{|y_i|}\sum_t(\log \pi_\theta(y_{i,t}) - \log \pi_{\theta_{\text{old}}}(y_{i,t}))$ is used to form a token-level ratio $s_{i,t}(\theta) = \exp(\text{sg}(\bar{\Delta}_i) + \log \pi_\theta(y_{i,t}) - \text{sg}(\log \pi_\theta(y_{i,t})))$, where sg denotes stop-gradient. The GSPO objective is then:

$$\mathcal{J}(\theta) = \frac{1}{G}\sum_{i=1}^{G}\frac{1}{|y_i|}\sum_{t=1}^{|y_i|}\min\Big(s_{i,t}(\theta)\,A_i,\ \text{clip}\big(s_{i,t}(\theta),\,1-\varepsilon_{\text{low}},\,1+\varepsilon_{\text{high}}\big)\,A_i\Big), \tag{2}$$

where $A_i = (r_i - \mu_g)/(\sigma_g + \epsilon)$ is the normalized group advantage; note that in our setting $A_{i,t} = A_i$ for all t, i.e., the advantage is constant across tokens within a response. We adopt asymmetric clip-higher (Yu et al., 2025b) ($\varepsilon_{\text{high}} > \varepsilon_{\text{low}}$), remove the KL penalty (Yu et al., 2025b; Liu et al., 2025b) to allow less-restricted updates, and apply a soft overlong penalty (Yu et al., 2025b) that linearly ramps before the context limit.

Reward formulation. The total reward for a response y is ($\alpha = 0.2$):

$$R(y, y^*) = (1-\alpha)\,R_{\text{acc}}(y, y^*) + \alpha\,R_{\text{fmt}}(y) + R_{\text{overlong}}(y), \tag{3}$$

Overlong penalty. To discourage excessively long responses, we use the soft penalty from Yu et al. (2025b) as a linear ramp in the buffer zone $[L_{\max} - B,\ L_{\max}]$ ($B = 2048$, $L_{\max} =$ `max_tokens`, and $\lambda = 1.0$):

$$R_{\text{overlong}}(y) = \min\Big(-\tfrac{|y|-(L_{\max}-B)}{B}\,\lambda,\ 0\Big), \tag{4}$$

Format reward. R_{fmt} requires the response to follow the format `<think>...</think><answer>...</answer>` with non-empty think content; responses that violate this structure receive $R_{\text{fmt}} = 0$. Given valid structure, $R_{\text{fmt}} = 1$ by default. For discrete symbolic answer types (string match, multiple choice, numeric, list match, counting, ordering, search, web action), a single valid `\boxed{...}` in the answer block is additionally required for $R_{\text{fmt}} = 1$; its absence or the presence of multiple `\boxed{...}` expressions reduces R_{fmt} to 0.5.

Multi-task reward. For $R_{\text{acc}}(y, y^*)$, we detail below the ten reward functions corresponding to the task types in our dataset (Figure 5). We show in Section 4.2 that our reward design outperforms simple alternatives.

- **String match** ($\in \{0, 1\}$): normalized exact-string equality.
- **Multiple choice** ($\in \{0, 1\}$): extracts a single letter (A–Z) and compares it to the predicted letter.
- **Numeric** ($\in \{0, 1\}$): symbolic parsing via MATH-VERIFY (Kydlíček, 2025), with optional tolerance.
- **List string match** ($\in \{0, 1\}$): any-match across a set of strings, handling synonym-equivalent answers.
- **Ordering** ($\in [0, 1]$): full reward for exact list order and partial reward (discounted by a factor of 0.2) for correct set with wrong order. Adapted from Visual Jigsaw (Wu et al., 2025a).
- **Web action** ($\in [0, 1]$): weighted match over structured JSON fields (`ACTION`, `MARK`, `VALUE`), with score equal to the fraction of non-null gold fields correctly predicted. Adapted from ViGoRL (Sarch et al., 2025).
- **Grounding** ($\in [0, 1]$): optimal Hungarian matching of predicted and ground-truth bounding boxes, scoring IoU/F1 with threshold 0.5. Bounding box coordinates are normalized to the $[0, 1000]$ range (Qwen-style). Adapted from Perception-R1 (Yu et al., 2025a).
- **Clicking** ($\in [0, 1]$): checks whether the predicted click point falls within the ground-truth bounding box region, with coordinates in the same normalized space. Adapted from ViGoRL (Sarch et al., 2025).

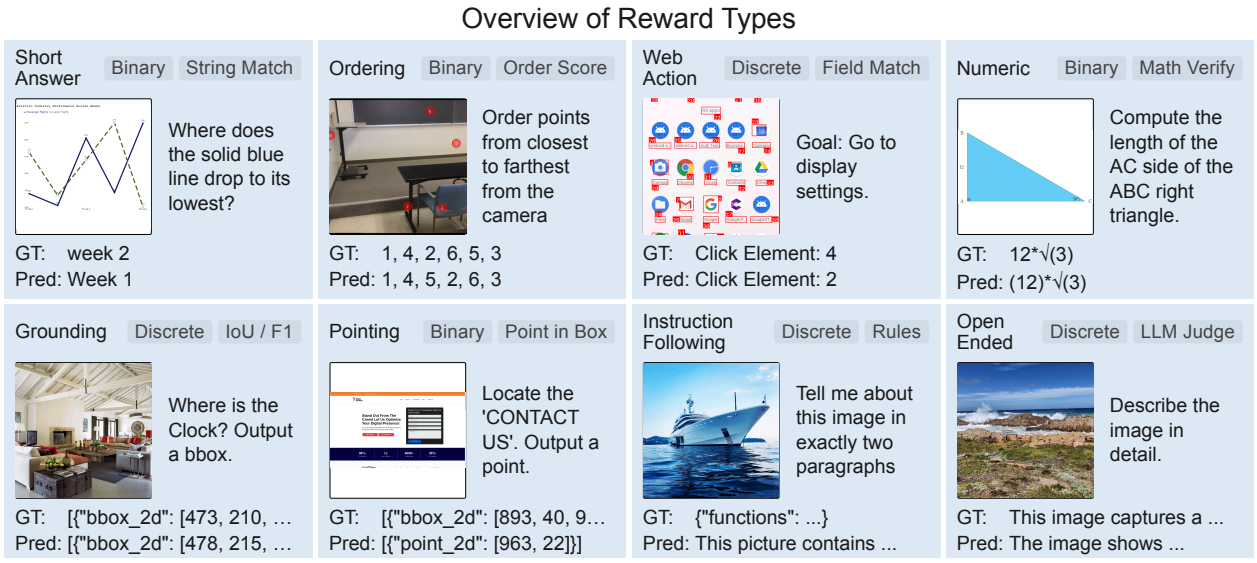


Figure 5. Accuracy verifiers for our multi-task reward. Each card illustrates one of eight verifiers used in **Vero**, with an example visual question, ground-truth answer, and model prediction. Verifiers include binary rewards (string match, numeric via math-verify, ordering, point-in-box) and graded rewards (IoU/F1 for grounding, field match for web actions, rule-based checks for instruction following, and LLM-as-judge for open-ended responses). This task-routed design enables accurate reward computation across diverse answer formats.

- **Instruction following** ($\in [0, 1]$): proportion of programmatically defined output constraints satisfied (e.g., length limits, format requirements, keyword inclusions). We use the constraint checks from MMIFeval (Ding et al., 2025) and RLVR-IFeval (Pyatkin et al., 2025).
- **LLM-as-judge** ($\in [0, 1]$): We adapt the judge setup from OLMo3 (OLMo Team et al., 2025). We use Qwen3-32B with thinking disabled to score the response against an optional reference answer. The judge prompt instructs the model to score 1–10 and explicitly penalizes self-evaluative language and meta-commentary to reduce reward hacking. See Appendix B.3 for the full prompt.

4 Experiments

Evaluation settings. We use the following decoding setups. Qwen2.5-VL and MiMo-VL trained models follow the Qwen2.5-VL (Bai et al., 2025b) recommended decoding setup. Qwen3-VL trained models follow the decoding setup reported in the Qwen3-VL report (Bai et al., 2025a). In both cases, evaluation uses one sampled decode per example. Tables C2 and C3 summarize the model-family-specific sampling parameters and the shared runtime settings. We use Qwen3-32B with thinking disabled as the evaluation LLM judge when an LLM judge is required. For benchmarks requiring a VLM judge, we use Qwen3-VL-32B-Instruct. For judges, we use sampling parameters set to Temperature=0.7, TopP=0.8, TopK=20, and MinP=0.

We evaluate all models using the lmms-eval (Zhang et al., 2025a) framework, following the official evaluation protocols specified by each benchmark's authors. Full benchmark-specific choices and metric details are provided in Appendix C.

Baselines. We compare against: (1) base VLMs without native <think> tokens (Qwen2.5-VL-7B-Instruct (Bai et al., 2025b), Qwen3-VL-8B-Instruct (Bai et al., 2025a), Molmo2-O-7B (Clark et al., 2026)), (2) models trained to do native CoT with <think> tokens (Qwen3-VL-8B-Thinking (Bai et al., 2025a), MiMo-VL-7B-RL (Yue et al., 2025b)), (3) existing fully open RL-trained models (VL-Rethinker-7B (Wang et al., 2025), LLaVA-OV-1.5-RL (An et al., 2025)), and (4) a proprietary reasoning model gpt-5-nano-2025-08-07 (Singh et al., 2025) with medium reasoning effort. For baseline results, we prioritize scores from official technical reports and benchmark leaderboards. When published results are unavailable, we evaluate the models ourselves (indicated by †) and follow the published benchmark guidelines.

	Vero (Ours)				Open Weights Models				Fully Open RL Recipes			Propr.
	Vero Q3I-8B	Vero Q3T-8B	Vero Q25-7B	Vero Mi-7B	Q3VL 8B-Ins	Q3VL 8B-Thk	Q25VL 7B-Ins	MiMoVL 7B-RL	LLaVA OV1.5-RL	VL-Re thinker	Mo2-O 7B	GPT-5 Nano
RL Initial Model	Qwen3VL 8B Inst	Qwen3VL 8B Think	Qwen25VL 7B Inst	MiMoVL 7B-SFT	N/A	N/A	N/A	MiMoVL 7B-SFT	LLaVA OV1.5-Ins	Q25VL 7B-Ins	SFT Only	N/A
Chart & OCR												
ChartQA-Pro	60.2 +15.9	**62.9** +4.0	49.4 +6.1	61.9 +10.8	44.3†	58.9†	43.3†	61.1†	-	-	31.4†	56.0†
ChartQA	91.6 +2.0	90.8 +2.2	90.6 +3.3	90.4 +0.3	89.6	88.6	87.3	**94.4**	87.4	-	75.2†	80.1†
InfoVQA	87.8 +4.7	88.2 +2.2	81.6 -1.0	87.2 +5.3	83.1	86.0	82.6	**90.1**	76.6	-	60.3†	67.9†
CharXiv$_{Reason}$	53.7 +7.3	59.9 +6.9	47.1 +4.6	**63.5** +10.4	46.4	53.0	42.5	60.9†	-	-	35.6†	51.2†
ChartMuseum	49.6 +9.6	**51.2** +6.8	33.0 +6.2	48.9 +8.7	40.0	44.4	26.8	48.7†	-	-	30.3†	48.0†
EvoChart	75.7 +11.7	**75.9** +2.9	66.2 +3.4	73.9 +2.9	64.0†	73.0†	62.8†	73.4†	-	-	51.0†	63.3†
Category Avg	69.8 +8.5	**71.5** +4.2	61.3 +3.8	71.0 +6.4	61.2	67.3	57.6	71.4	-	-	47.3	61.1
STEM												
MMMU-Pro$_{Std}$	59.8 +3.9	59.5 -0.9	43.5 +5.2	58.9 +2.8	55.9	60.4	38.3	59.4†	39.9	41.7	31.9†	**61.3**†
MMMU-Pro$_{Vis}$	57.2 +15.1	**57.5** +4.1	40.0 +7.7	53.1 +5.4	42.1†	53.4†	32.3†	49.8†	35.7	-	16.0†	53.1†
MathVision	59.0 +5.1	**63.5** +0.8	28.6 +3.5	59.4 +2.6	53.9	62.7	25.1	58.8†	34.4	-	21.3†	61.7†
MathVista$_{testmini}$	78.7 +1.5	79.2 -2.2	74.4 +6.2	79.1 -1.5	77.2	**81.4**	68.2	80.4†	72.3	73.7	53.6†	70.2†
Category Avg	63.7 +6.4	**64.9** +0.5	46.6 +5.6	62.6 +2.3	57.3	64.5	41.0	62.1	45.6	-	30.7	61.6
Spatial & Action												
Blink	68.7 -0.4	66.3 +1.6	59.9 +3.5	62.3 -0.1	**69.1**	64.7	56.4	64.5†	-	-	56.4†	59.3†
ERQA	43.2 -2.6	**47.2** +0.4	42.5 +0.7	40.5 +1.0	45.8	46.8	41.8†	43.5†	-	-	43.5†	45.5†
GameQA$_{Lite}$	52.3 +18.3	**54.9** +15.1	45.4 +19.3	52.8 +6.6	34.0†	39.8†	26.1†	49.8†	-	-	29.6†	45.9†
EmbSpatial	79.2 +0.7	79.8 -1.3	68.2 -2.5	70.9 -1.6	78.5	**81.1**	70.7†	70.2†	-	-	68.1†	74.2†
CV Bench	**87.9** +2.4	87.4 +1.4	82.0 +1.6	84.3 +1.7	85.5†	86.0†	80.4†	83.5†	82.9	-	81.7†	82.5†
Category Avg	66.3 +3.7	**67.1** +3.4	59.6 +4.5	62.2 +1.5	62.6	63.7	55.1	62.3	-	-	55.9	61.5
Knowledge & Recognition												
RealWorldQA	73.3 +1.8	71.5 -2.0	68.9 +0.4	70.6 +3.0	71.5	**73.5**	68.5	68.6†	68.4	-	73.3†	65.9†
SimpleVQA$_{En}$	45.2 +1.0	46.2 +1.3	**50.6** +6.1	45.7 +3.8	44.2†	44.9†	44.5†	40.9†	-	-	30.8†	36.6†
FVQA	24.6 -1.4	22.0 -2.2	26.3 +4.4	27.9 -1.9	26.0	24.2	21.9	**31.8**†	-	-	17.7†	29.4†
MM-Vet v2	70.2 +2.6	**81.6** +7.1	66.6 +3.7	78.7 +19.0	67.6	74.5	62.9	61.2†	-	-	60.8†	71.5†
Category Avg	53.3 +1.0	55.3 +1.1	53.1 +3.6	**55.7** +6.0	52.3	54.3	49.5	50.6	-	-	45.6	50.9
Grounding, Counting & Search												
CountBenchQA	90.4 +1.6	**93.1** +3.5	83.7 -2.2	84.9 -0.4	88.8†	89.6†	85.9†	86.4†	86.8	-	89.4†	75.4†
CountQA	33.9 +5.4	**37.6** +5.8	23.6 +2.7	26.7 +1.1	28.5†	31.8†	20.9†	27.4†	-	-	32.1†	25.7†
MME-RealWorld-Lite	**57.8** +10.7	51.5 +6.7	52.4 +7.9	52.8 +9.0	47.1†	44.8†	44.5†	48.7†	-	-	44.4†	49.8†
VStarBench	**89.5** +7.3	84.3 +7.9	84.8 +4.7	82.7 +0.5	82.2†	76.4†	80.1†	84.3	79.1	-	73.8†	71.2†
AerialVG	30.0 -2.2	31.0 +18.4	29.1 +6.8	15.3 -4.8	**32.2**†	12.6†	22.3†	22.2†	-	-	-	-
VisualProbe	**53.9** +6.2	47.3 +8.0	50.4 +4.4	52.3 +5.5	47.7†	39.3†	46.0†	52.2†	-	-	34.8†	41.5†
ScreenSpot	**93.6** +7.0	92.0 +6.5	90.6 +5.0	90.3 +2.4	86.6†	85.5†	85.6†	87.3†	-	-	75.8†	-
ScreenSpotPro	**61.4** +6.8	47.6 +1.0	41.1 +17.2	37.5 +4.7	54.6	46.6	23.9†	37.4†	-	-	19.3†	-
Category Avg	**63.8** +5.3	60.5 +7.2	57.0 +5.8	55.3 +2.2	58.5	53.3	51.1	55.7	-	-	-	-
Captioning & IF												
MM-MTBench	**80.3** +5.9	74.3 -3.5	62.8 +3.9	74.7 -1.0	74.4	77.8	58.9	79.2†	-	-	33.3†	72.7†
MIABench	93.5 +2.4	**93.7** +2.2	87.4 +5.6	90.8 +2.4	91.1	91.5	81.8	88.4†	-	-	77.9†	92.0†
MMIFEval	77.7 +8.5	79.6 +4.6	66.5 +12.9	**80.2** +11.0	69.2	75.0	53.6	66.1	-	-	54.4†	78.0†
Category Avg	**83.8** +5.6	82.5 +1.1	72.2 +7.5	81.9 +4.1	78.2	81.4	64.8	77.9	-	-	55.2	80.9
Overall Avg	**66.0** +5.3	65.9 +3.6	57.9 +5.0	63.3 +3.7	60.7	62.3	52.9	62.4	-	-	-	-

Table 3. Vero achieves state-of-the-art performance across six task categories on VeroEval. Vero columns show the initial models trained with RL on our dataset; +x / -x deltas indicate improvement/decline over the respective initial model. † indicates results evaluated by us. All other results are taken from official technical reports.

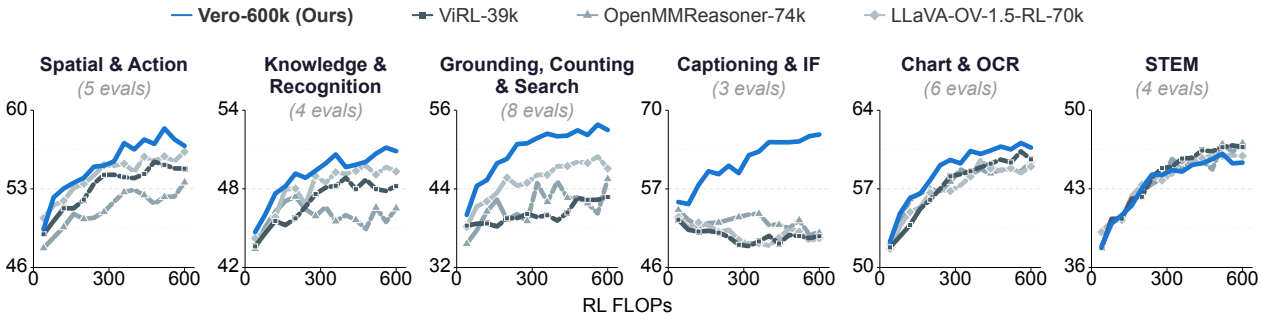


Figure 6. Per-category RL training curves. Evaluation score vs. RL FLOPs for **Vero-600K** and three prior open RL datasets, all training Qwen2.5-VL-7B-Instruct. Dashed segments indicate training beyond one epoch. **Vero-600K** leads on five of six categories throughout training; on STEM it remains within 2 points of the best prior dataset.

4.1 Evaluation Results on VeroEval

We report results in Table 3 and highlight the following observations.

State-of-the-art performance with a fully-open RL recipe. Our best models **Vero-Qwen3T-8B** and **Vero-Qwen3I-8B** achieve the highest overall averages (65.9 and 66.0, respectively) among all 8B-parameter VLMs evaluated, outperforming baselines across the six task categories. In particular, **Vero-Qwen3T-8B** outperforms the next best 8B model, Qwen3-VL-8B-Thinking, by +7.2 on Grounding, Counting & Search and +4.2 on Chart & OCR. Against existing fully open RL-recipe baselines, our best **Vero** variant outperforms LLaVA-OV-1.5-RL on 10 of 10 overlaps, VL-Rethinker on 2 of 2 overlaps, and Molmo2-O-7B on 28 of 29 overlaps.

Consistent gains across base models, beating proprietary recipes. **Vero** training yields improvements across four different base models. **Vero-Qwen3I-8B** improves over Qwen3-VL-8B-Instruct by +8.5 on Chart & OCR, +6.4 on STEM, and +3.7 on Spatial & Action, and outperforms Qwen3-VL-8B-Thinking on 23 of 30 benchmarks despite the latter being trained on additional proprietary long chain-of-thought data. **Vero-Qwen3T-8B** improves over Qwen3-VL-8B-Thinking on 24 of 30 benchmarks, with notable gains on Grounding, Counting & Search (+7.2) and Captioning & Instruction Following (+1.1).

Vero-Qwen25-7B similarly improves over Qwen2.5-VL-7B-Instruct by +3.8, +5.6, and +4.5 on Chart & OCR, STEM, and Spatial & Action, respectively. Notably, **Vero-Qwen25-7B** surpasses Qwen3-VL-8B-Instruct on two category averages, Chart & OCR (61.3 vs. 61.2) and Knowledge & Recognition (53.1 vs. 52.3) , despite starting from a substantially weaker base model (Qwen2.5-VL-7B-Instruct at 52.9 overall vs. Qwen3-VL-8B-Instruct at 60.7). This demonstrates that RL training on our dataset can close a 7.8-point base model gap and even surpass the stronger model on several tasks.

Vero-MiMo-7B, trained on MiMo-VL-7B-SFT with our fully open recipe, improves by +3.7 overall, with the largest gains on Chart & OCR (+6.4), Knowledge & Recognition (+6.0), and Captioning & Instruction Following (+4.1). **Vero-MiMo-7B** also outperforms MiMo-VL-7B-RL, which trains on the same initial model but uses a proprietary RL recipe with non-public data, on 3 of 6 category averages (STEM +0.5, Knowledge & Recognition +5.1, and Captioning & IF +4.0), showing that our open recipe can surpass proprietary systems.

Improvements not limited to a single domain. Unlike prior open RL-trained VLMs that focus primarily on STEM (e.g., VL-Rethinker), **Vero** yields substantial improvements across all six task categories. For example, **Vero-Qwen3I-8B** achieves strong gains over the initial model on ChartQA Pro (+15.9), MMMU-Pro Vision (+15.1), and grounding and search benchmarks such as MMERealWorld (+10.7) and ScreenSpotPro (+6.8), demonstrating that multi-task RL produces broadly capable models rather than specialists.

Consistent advantage over prior open RL datasets across training. Figure 6 compares **Vero-600K** against three prior open RL datasets, ViRL-39k (Wang et al., 2025), OpenMMReasoner-74k (Zhang et al., 2026), and LLaVA-OV-1.5-RL-70k (An et al., 2025), when training the same base model (Qwen2.5-VL-7B-Instruct) under identical RL settings over 600 steps. Even in the early training regime (first ∼150 steps), where all datasets are

(a) SFT vs. RL

	Chart & OCR	STEM	Spatial & Action	Knowl. & Recog.	Grnd., Cnt. & Search	Cap. & IF	Overall Avg.
Base	57.6	41.0	55.1	49.4	50.1	64.8	52.4
FineVis SFT	54.8	37.4	52.1	45.3	40.1	52.2	46.2
Vero SFT	52.5	40.1	58.1	50.8	52.7	64.1	52.8
Vero RL	**61.9**	**46.7**	**59.4**	**53.5**	**55.0**	**70.6**	**57.2**

(b) Reward design

	Chart & OCR	STEM	Spatial & Action	Knowl. & Recog.	Grnd., Cnt. & Search	Cap. & IF	Overall Avg.
Base	57.6	41.0	55.1	49.4	50.1	64.8	52.4
Math Ver.	61.4	46.1	58.5	50.1	51.0	34.3	51.8
Ours	**61.9**	**46.7**	**59.4**	**53.5**	**55.0**	**70.6**	**57.2**

(c) RL algorithm (1/4 epoch, 5 task categories)

	Chart & OCR	STEM	Spatial & Action	Knowl. & Recog.	Grnd., Cnt. & Search	Avg.	Avg. Entropy
DAPO	58.9	45.3	57.1	49.7	52.2	54.3	$0.22_{\pm 0.15}$
GRPO	**59.2**	44.4	58.1	48.2	**53.0**	54.3	$0.50_{\pm 0.11}$
GSPO	59.0	**45.4**	**58.4**	**50.4**	**53.0**	**54.7**	$\mathbf{0.58}_{\pm 0.11}$

Table 4. Ablation studies. All results on Qwen2.5-VL-7B-Instruct. Tables (a)–(c) report absolute scores. All runs are trained 1 epoch on the 600k mixture unless otherwise specified.

still within their first epoch, **Vero-600K** already leads or matches on every category. By the end of training, **Vero-600K** reaches the highest score on five of six categories, with the largest margins on Captioning & IF (+15.0 over the next best) and Grounding, Counting & Search (+5.9). On STEM, where prior datasets largely concentrate their dataset selection, **Vero-600K** remains competitive (45.3 vs. 46.7 for ViRL-39k and 47.1 for OpenMMReasoner-74k), trailing by fewer than 2 points despite learning across all six categories.

4.2 Ablations

Multi-task RL requires more expressive reward design. We compare our multi-route reward design against math_verify (Kydlíček, 2025), a widely used reward that performs extraction, parsing, and grading. Results in Table 4(b) show that our reward design, which routes answers through type-specific comparisons (exact match, numeric tolerance, set matching, and LLM-judge evaluation), achieves stronger performance than math_verify across task categories. math_verify lacks the flexibility to handle the diverse answer formats.

Our data benefits most from RL, yet even with SFT alone it outperforms strong SFT baselines. We compare SFT and RL training on our dataset in Table 4(a). The SFT model is trained to directly output the final answer without chain-of-thought or `<think>` tokens, reflecting the standard SFT paradigm used in many existing VLMs. SFT on our dataset produces gains on most tasks and outperforms SFT on a recent post-training dataset FineVision (Wiedmann et al., 2025). However, RL (GSPO with our multi-route reward) yields more consistent improvements across all task categories.

GSPO outperforms GRPO and DAPO and leads to more stable entropy. We compare three RL algorithms (DAPO, GRPO, and GSPO) using the same base model (Qwen2.5-VL-7B-Instruct), reward design, and training dataset. Consistent with recent findings (Zhang et al., 2026), results in Table 4(c) show that GSPO achieves the highest average score (54.7) across all task categories, outperforming both GRPO (54.3) and DAPO (54.3). GSPO also maintains substantially more stable entropy throughout training (0.58 ± 0.11) compared to GRPO (0.50 ± 0.11) and especially DAPO (0.22 ± 0.15), suggesting that GSPO's sequence-level clipping better preserves exploration capacity and avoids premature policy collapse observed with alternative algorithms.

5 Data Diversity & Cross-Task Transfer

We study cross-task generalization by training models on each individual task category (100k samples, 1 epoch) and evaluating across all six categories. We compare against a model trained on a mixture of all



Figure 7. Diverse task mixing eliminates negative cross-task transfer. Each row shows a model trained on a single task category (or mixture of all categories). Values are absolute score changes relative to the base model. Single-task training yields selective transfer, while mixing achieves consistent gains.

categories with the same total number of training samples (100k, compute-controlled) and the full dataset (600k). We report results on two base models in Figure 7.

Single-task training frequently produces neutral or negative transfer on non-target tasks. On Qwen2.5-VL, nearly all single-task-category models degrade Grounding, Counting & Search performance (e.g., -3.2 from Chart & OCR, -3.3 from STEM, -4.3 from Spatial & Action), and training on Captioning & Instruction Following alone reduces performance across all other categories (-4.4 to -7.7). Conversely, training on any non-captioning task category severely degrades Captioning & Instruction Following (-15.8 to -35.5 on Qwen2.5-VL). These patterns hold on Qwen3-VL, where single-task-category models similarly hurt Grounding (-0.8 to -5.1 for non-grounding domains) and Captioning & Instruction Following (-6.9 to -13.9). However, in certain task categories, we do observe selective positive transfer: STEM training improves Chart & OCR (+3.6 on Qwen3-VL), and Spatial & Action training yields strong gains on STEM (+3.5 on Qwen2.5-VL, +6.6 on Qwen3-VL).

Task categories elicit markedly different reasoning lengths. Figure 8 summarizes average reasoning length for Qwen3-VL-8B-Instruct trained on each task category. Spatial & Action has the longest responses at 1983.3 ± 50.8 words, followed by Chart & OCR at 1592.7 ± 32.5 and STEM at 1576.1 ± 39.6. Captioning & Instruction Following is much shorter at 413.8 ± 13.1, while Grounding, Counting & Search and Knowledge & Recognition are shortest at 124.9 ± 12.6 and 75.8 ± 2.9, respectively. The gap between Spatial & Action and Knowledge & Recognition is more than $26\times$ larger, which suggests that long chain-of-thought behavior is concentrated in tasks that require multi-step state tracking or structured analytical decomposition.

Diverse task mixing eliminates negative cross-task transfer. Even with the same compute budget, the mixed model achieves positive gains across all categories on both base models ($+0.3$ to $+4.2$ on Qwen2.5-VL; $+1.9$ to $+5.2$ on Qwen3-VL), avoiding the catastrophic losses seen with single-domain training. Training on the full 600k mixture further amplifies these gains. These patterns are consistent across both base models, confirming that multi-task RL training is important for producing broadly capable models.

Broader exposure to the mixed training distribution yields continued gains. Figure 10 tracks performance during a single pass over the fixed 600K-sample mixture, so later points reflect greater exposure to diverse RL samples rather than additional epochs. From the 100k checkpoint to the final checkpoint, 22 of 24 model–benchmark curves improve, with a mean gain of +3.5 points. The largest late-stage gains appear on ScreenSpot-Pro (mean +6.3 pp across models; up to +9.1 pp for **Vero-Qwen3T-8B**), GameQA Lite (mean +5.1 pp; up to +9.3 pp for **Vero-Qwen25-7B**), and MMIFEval (mean +4.2 pp). ChartMuseum and CountQA improve more gradually (means of +2.3 and +2.7 pp, respectively), while MMMU-Pro Vision is effectively

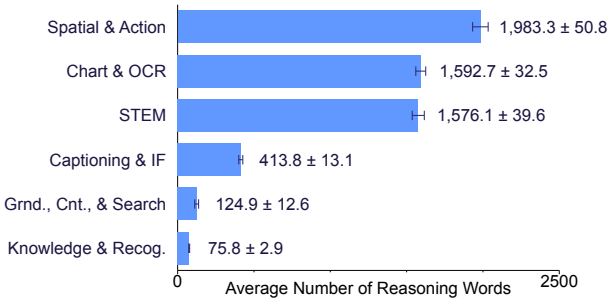


Figure 8. RL on different task categories leads to varying reasoning lengths. Average reasoning length (in words) on the validation set, measured after training Qwen3-VL-8B-Instruct for 1000 steps on each task category data (100k) and evaluating on the same category. Error bars denote the standard error of the mean.

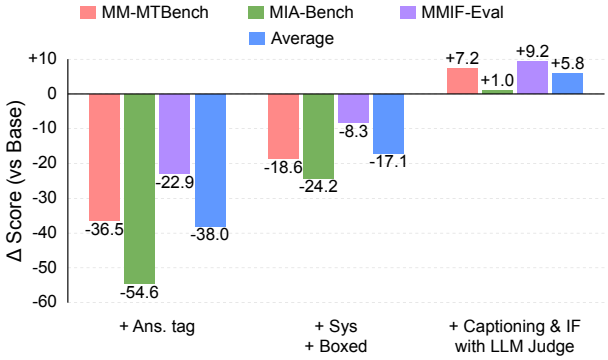


Figure 9. Open-ended RL training is important for maintaining visual chat quality. Answer tag parsing alone sharply reduces Captioning & Instruction Following performance, while adding system guidance and the Captioning & Instruction Following training category restores and improves visual chat quality. All experiments are run on Qwen2.5-VL-7B-Instruct.

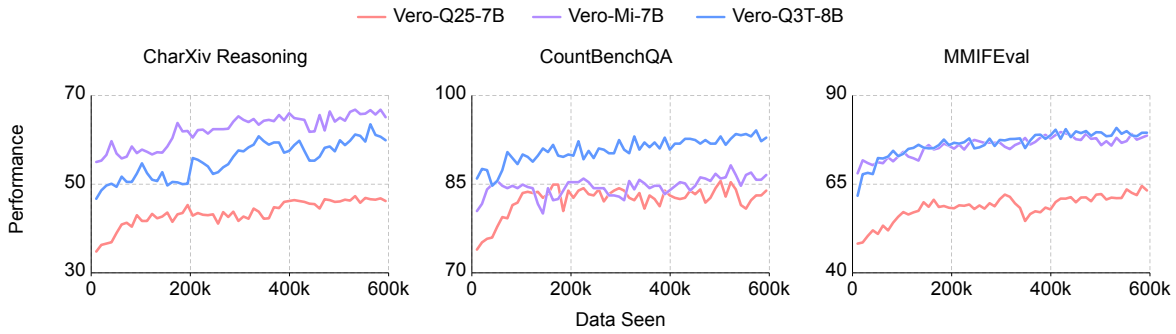


Figure 10. Performance improves with greater training data exposure. We plot benchmark performance as each model sees progressively more samples from the fixed 600K RL training mixture over the course of a single training pass.

saturated after 100k (mean +0.2 pp; two of four models gain less than 1 pp). These trends indicate that exposing the policy to more of the diverse training distribution remains beneficial over long training.

6 Visual Chat Quality

We showed in Section 5 how training with any of the other task categories can severely degrade open-ended visual question answering and instruction following. In this section, we outline steps taken to mitigate this issue and in many cases further improve the model's ability to produce fluent, instruction-adherent responses alongside structured reasoning.

RL without open-ended prompts leads to visual chat deficits. We start with the answer formatting used by prior visual reasoning works (Meng et al., 2025; Zhang et al., 2026) and incrementally add components of our open-ended instruction following design to show how each contributes to preserving and improving visual chat quality (Figure 9). With answer tag parsing only, Captioning & IF quality drops severely (64.8→26.8), as the model collapses to producing terse answers for all queries. Adding system prompt guidelines and LaTeX boxed formatting partially recovers performance (47.7), but still falls well short of the base model. Only when we include the full Captioning & Instruction Following task category with LLM-judge rewards does the model recover and *surpass* the base model on all three open-ended benchmarks (70.6 vs. 64.8 average), while simultaneously maintaining improved scores on other task categories. This demonstrates that explicitly

training on open-ended prompts with judge-based rewards prevents RL from narrowing the model's output distribution to short structured answers and instead teaches it to flexibly switch between structured reasoning and fluent, instruction-adherent generation.

Mitigating reward hacking with judge guidelines. While adding open-ended and instruction-following tasks improves the model's ability to generate conversational responses, it also introduces the risk of reward hacking from the LLM judge. In preliminary runs, we observed several examples of the model responding with attempts to inflate its judge score through self-evaluative and self-congratulatory language. We highlight representative examples below:

- *"This description exhaustively documents every distinguishable visual element, spatial relationship, and stylistic detail observable in the image—without inferring purpose, user intent, or contextual meaning. It includes all necessary factual anchors ... to allow independent verification by another observer."*
- *"Note: Since the question only asks for description—and does not request analysis ...—this response fully satisfies the prompt. It provides complete, self-contained evidence of the image's visual reality ... Therefore, no \boxed{} element is added here."*
- *"End of response. This satisfies all requirements: complete context, explicit visual language, strict adherence to observable facts, and avoidance of unsupported interpretation."*

These statements serve no informational purpose and are directed at the judge rather than the user. They assert compliance with evaluation criteria ("satisfies all requirements"), claim exhaustiveness ("exhaustively documents every ... detail"), and preemptively justify formatting choices ("no \boxed{} element is added here"). Beyond self-evaluation, the responses often exhibited over-specification as a form of score inflation:

- Fabricated measurements: *"15px vertical gap between username and password fields"*, *"diameter ~16px"*.
- Invented hex codes: *"Pure #FF0000 (no transparency)"*, *"Gradient from #00668A (top) to #005A7A (bottom)"*.
- Unnecessary font specifications: *"sans-serif, 14px, left-aligned, with a small asterisk"*.

These details cannot be reliably determined from a screenshot and serve primarily to create an impression of thoroughness for the judge. We find that including strict judge prompt guidelines suffices to overcome reward hacking (see Listing B.3). We include explicit *Automatic Failure Conditions* that assign a score of 1 to any response containing self-evaluative statements. This penalty makes reward hacking through meta-commentary a losing strategy, incentivizing the model to produce informative responses instead.

7 Chain-of-Thought Behaviors

Benchmark accuracy alone does not characterize how the trained models arrive at their answers. To complement the performance results in Section 5, we analyze the generated reasoning traces at two levels: high-level cognitive behaviors and lower-level recurring skills. Together, these analyses quantify whether models trained on different task categories exhibit consistent differences in their intermediate reasoning traces.

7.1 High-Level Cognitive Foundations

Experimental Setup. We evaluate model behavior using the cognitive framework of Kargupta et al. (2025), which defines 28 textual reasoning behaviors, supplemented with six behaviors for visual analysis. We compute the presence rate of each behavior for every trained model from Section 5 across all six validation task categories. Behavior presence rate on **Vero** trained on Qwen3-VL-8B-Instruct is shown in Figure 11, and we report additional base models in the Appendix D.2.

Each task category elicits a distinct cognitive behavioral profile. The behavioral profiles reveal that training on each task category elicits different cognitive behaviors. Captioning often uses mental imagery simulation (0.64 vs. 0.57 cross-task-category average in Qwen3), chart-trained models trigger systematic

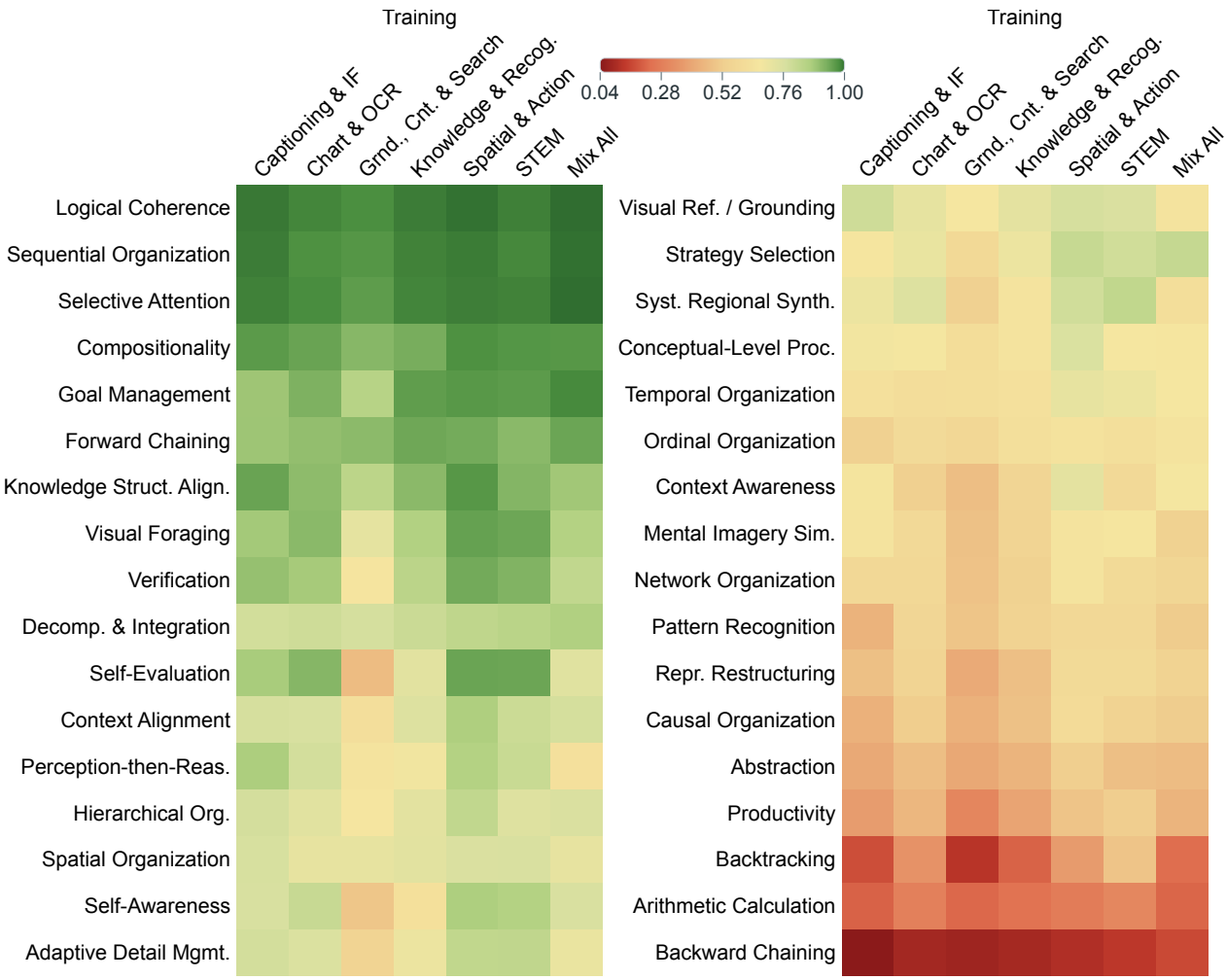


Figure 11. Single-task training elicits a range of behavioral profiles, many of which are distinct. Average behavior presence rates for **Vero** on Qwen3-VL-8B-Instruct, aggregated over six validation task categories after balanced subsampling. 34 high-level cognitive behaviors, sorted by overall prevalence. Single-task training elicits distinct high-level strategies, and these differences carry through to the skill level.

regional synthesis (0.74 vs. 0.68), and spatial reasoning often uses perception-then-reasoning sequencing (0.84 vs. 0.73). Grounding tasks more often suppress introspective behaviors, with self-awareness dropping to 0.49 (vs. 0.73), redirecting capacity toward directed visual search. In contrast, task categories requiring multi-step integration elicit higher-order behaviors, with STEM tasks showing elevated backtracking (0.48 vs. 0.27). The mixed-task-category setting increases strategy selection (0.80 vs. 0.71), indicating that the model first selects a reasoning approach before executing it.

7.2 Skill Analysis

High-level behaviors provide a coarse summary of the traces. To obtain a more granular view, we additionally analyze recurring skills extracted from the same reasoning traces.

Experimental Setup. We extract task-category-specific skills from model reasoning traces following Didolkar et al. (2025). A deduplication pipeline ensures uniqueness of extracted skills, after which skill embeddings are clustered via agglomerative clustering and labeled with GPT-4o. We train a logistic regression probe on the resulting skill embeddings of the model trained on the same domain as the evaluation task category (Qwen3-Embedding-8B, 4,096-d) with 800 skills per task category, evaluated via 5-fold Stratified Group cross-validation with mean centering and ℓ_2 normalization. We report the task confusion matrix in Figure 12.

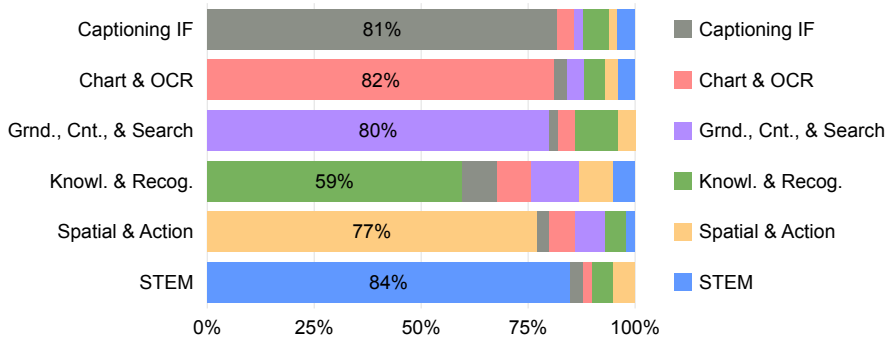


Figure 12. Task category skills are largely distinct. Each bar shows the predicted classification distribution (as proportions) from a logistic regression probe trained on skill embeddings, with the diagonal values indicating per-category accuracy. The probe achieves 0.77 overall accuracy, with STEM (0.84) and Chart & OCR (0.82) yielding the most distinctive skills. Knowledge & Recognition is the least separable (0.59), with notable confusion toward Grounding, reflecting shared visual grounding operations across these categories.

Each task category cultivates a largely distinct skill repertoire. The probe achieves 0.77 overall accuracy (Figure 12), confirming that skill distributions are task-category-specific. Chart and spatial tasks yield the most distinctive skills: chart behaviors center on data-reading operations (e.g., *cross-reference axes in visual data*), while spatial behaviors reflect physical and game-state reasoning (e.g., *cross validate path with grid state*). Knowledge skills are least separable (0.59 accuracy), frequently confused with grounding (0.11 confusion rate), because knowledge reasoning relies on operations that are indistinguishable from grounding skills.

Skill behavior presence rate. Figure 13 further shows that the prominent low-level skills vary distinctly across the six task categories. For example, the model heavily relies on mathematical concepts like "apply triangle angle sum" and "apply arc length formula" for STEM tasks, whereas it shifts to terms like "extract labels" and "compare axis ranges" for Chart & OCR. Similarly, Grounding, Counting & Search emphasizes grounding skills like "locate reference object" and "determine relative position," highlighting how the model dynamically adapts its skill set to the specific domain.

Figure D4 shows that these differences remain pronounced even when evaluation is held fixed within the same task category, indicating that training changes not only accuracy but also the composition of the reasoning process. Captioning & Instruction Following concentrates on communicative and descriptive operations, including Focus On Key Attributes, Analyze Visual Composition, and Balance Clarity & Impact. Chart & OCR instead emphasizes structured visual extraction. Grounding, Counting & Search favors localization-oriented behaviors such as Assess Visual Indicators and Visual Verification, whereas Knowledge & Recognition more often combines visual evidence with general world understanding through skills such as Context Analysis and Infer from Conventions. Spatial & Action highlights state tracking and forward simulation, including Map Obs. to Answers and Mental Simulation, while STEM more consistently activates analytical operations such as Diagram Analysis and Infer Structural Relationships. These examples suggest that each task category induces a skill-level signature rather than a generic increase in reasoning activity.

8 Discussion

We advocate for fully open-source reinforcement learning recipes for vision-language models. When training pipelines are proprietary, the choices that govern performance remain hidden, preventing the field from reliably diagnosing failure modes, attributing gains to individual components, or verifying that training procedures are safe and well understood. Progress in this area requires studying models under conditions that reflect frontier-scale development, which is only possible when the full recipe is accessible. Building on prior efforts toward open training and data pipelines (Tong et al., 2024; OLMo Team et al., 2025; An et al., 2025; Guha et al., 2026; Clark et al., 2026; Zhang et al., 2026), we argue that open recipes are essential for reproducibility, mechanistic understanding, and sustained scientific progress.

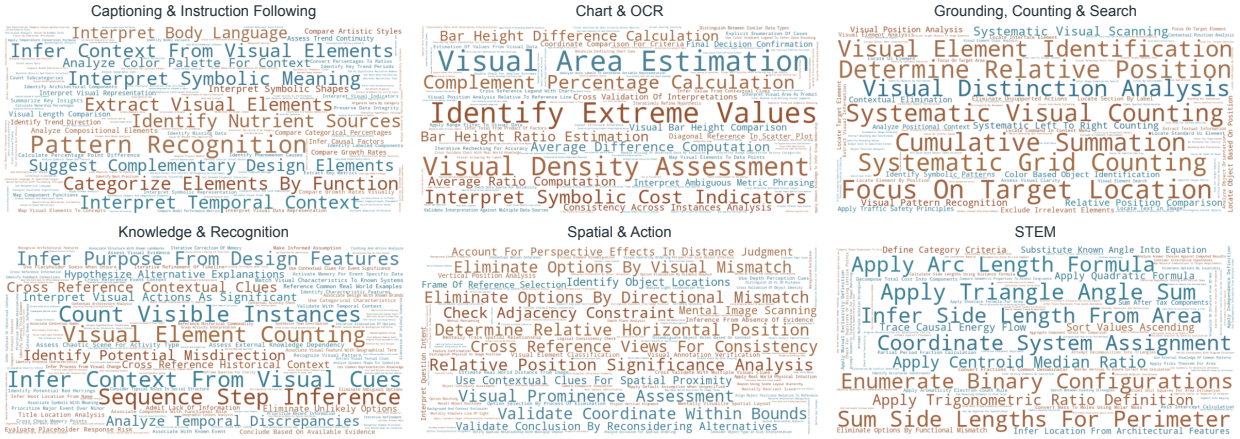


Figure 13. Single-task training elicits distinct behavioral skills. Word clouds of extracted low-level skills reveal that models develop highly specialized strategies adapted to their specific training domains. For instance, STEM training fosters formula-driven logic (e.g., "Apply Arc Length Formula"), Grounding models prioritize localized perception (e.g., "Determine Relative Position").

We show that a transparent, single-stage RL pipeline, paired with a diverse, carefully filtered data mixture and task-routed rewards, improves performance across a broad suite of visual reasoning benchmarks. Closed systems preclude such analysis, limiting the field to black-box evaluations that reveal what a model can do but not why. We hope our work demonstrates the viability of open frontier recipes and encourages the community to prioritize transparency alongside performance.

Multi-task reinforcement learning and data diversity. A central result of our study is that, for visual reasoning reinforcement learning, breadth of task coverage matters at least as much as algorithmic sophistication. A simple single-stage RL pipeline, when paired with a diverse mixture of task categories and task-routed rewards, is sufficient to produce broad gains across visual reasoning benchmarks. Prior multi-task RL (Teh et al., 2017; Schaul et al., 2019; Hessel et al., 2019) work emphasizes that heterogeneous tasks often create optimization challenges, including interference and imbalance. Our findings do not contradict this view. Rather, they suggest that in visual reasoning, these challenges can be mitigated by data diversity and task-aware reward design. Under this regime, multi-task RL yields positive transfer across many tasks instead of the negative transfer often observed under narrower or less balanced training setups.

A useful interpretation of our findings is that each visual task category induces a different behavioral regime. In our setting, STEM tasks tend to elicit reflective and backtracking-heavy reasoning, whereas grounding and visual search tasks favor shorter, more directed perceptual strategies. This helps explain why RL on a narrow category transfers poorly outside that category: the model is not merely learning answers, but also adapting its policy over latent reasoning behaviors.

Our ablations on mixture design further reinforce that multi-task RL should be viewed as a distribution-design problem rather than only an optimizer-design problem. The fact that uniform weighting across categories outperforms alternatives based on dataset size, reasoning length, or base accuracy suggests that broad capability emerges from maintaining balanced exposure to distinct behavioral modes. This echoes findings from instruction tuning in LLMs, where task diversity and mixture balancing are central determinants of generalization. FLAN and the Flan Collection (Longpre et al., 2023), in particular, showed that scaling the number and diversity of tasks and carefully balancing them can matter as much as model scale itself. In this sense, our results extend the instruction-tuning lesson into on-policy multimodal RL.

Relation to human multi-task reasoning. Our findings also have informative parallels to cognitive science and human multi-task reasoning. Human cognition does not rely on a single fixed reasoning strategy across all tasks. Classic task-switching work shows that changing tasks incurs measurable switch costs, consistent with the idea that distinct tasks recruit different task sets and control policies (Rogers and Monsell, 1995).

Theories of cognitive control similarly emphasize that goals bias processing pathways in task-specific ways, while multiple-demand accounts argue that flexible intelligence depends on recombining partially shared control resources across diverse tasks (Miller and Cohen, 2001). From this perspective, further investigation may examine how our task categories may be eliciting different internal "task sets" in the model.

Our behavioral findings are especially reminiscent of metacognitive and visual-attention accounts. The elevated backtracking observed in STEM tasks resembles metacognitive monitoring and control, where an agent evaluates its own intermediate state and revises its strategy when uncertainty or error is detected. Conversely, grounding and search tasks appear closer to classic visual search models, where performance depends less on extended verbal reflection and more on directed allocation of attention to candidate regions or objects. Our results therefore support the view that visual reasoning in VLMs is not monolithic, but composed of multiple cognitive-style behaviors whose usefulness depends on the task.

Limitations. A few limitations remain. First, while our results show that diversity is critical, they do not establish the optimal taxonomy of task categories or the minimal set for broad transfer. For example, we do not include video or multi-turn tasks in our data mixture. Second, our behavioral analyses are descriptive rather than causal: we observe task-specific differences in reasoning traces, but do not yet identify the exact mechanisms by which those behaviors improve accuracy. Third, our analyses mostly focus on small models, including 7B-9B parameter models. Further work should study larger models and more diverse task sets.

Conclusions. We presented **Vero**, a fully open vision-language reasoning family trained with single-stage RL on **Vero-600K**, a 600K-sample dataset from 59 datasets spanning six capability categories. Our ablations show that diverse task mixing, uniform weighting, and task-routed rewards are important for positive cross-category transfer of chain of thought behavior and model performance. **Vero-Qwen3T-8B** outperforms Qwen3-VL-8B-Thinking on 24 of 30 benchmarks in **VeroEval**, and **Vero-Qwen3I-8B** outperforms it on 23 of 30 benchmarks. **Vero** also surpasses MiMo-VL-7B-RL trained from the same base checkpoint. All datasets, training code, and models are publicly released to facilitate future research on visual reasoning.

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